



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO GOPALAKRISHNA AGENCY UNIT - II

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 27.05 km
Estimated Duration: 0.7 hours
Adjusted Duration (Heavy Vehicle): 0.8 hours
Start: (11.4381, 77.8734)
End: (11.244396, 77.875587)



Welcome to the Journey Risk Management Study

Route Overview

The journey from Sangagiri to Solasiramani covers approximately 27.05 kilometers. The route likely follows smaller state and local roads, which may be narrow or in varying states of maintenance. The typical travel time for heavy vehicles is around 0.66 hours under normal conditions.

Typical Weather Conditions and Related Hazards

Tamil Nadu generally experiences a tropical climate. The weather can include:

- **Summer (March to June):** High temperatures and potential dust, requiring heat management for vehicles.
- **Monsoon (July to September):** Heavy rains contributing to slippery roads and possible flooding, especially in poorly drained areas.
- **Post-Monsoon/Winter (October to February):** More stable but cooler weather, ideal for safe travel but with potential fog in early mornings.

Traffic Patterns

Traffic is typically lighter outside urban centers:

- **Peak Hours:** Primarily morning (8 AM - 10 AM) and evening (5 PM - 7 PM).
- **Congestion Prone Areas:** Small-market areas in towns may cause minor delays.

Road Quality and Infrastructure

- Expect varied road quality with potential for potholes and uneven patches, especially post-monsoon.
- Narrowness of roads may pose challenges for large trucks, necessitating cautious navigation, especially in populated areas.

Alternative Routes for Emergencies

- **Via SH 86:** This state highway could offer a quicker, better-maintained alternative depending on conditions. Availability of alternative routes depends on real-time conditions like floods or roadwork.

Local Regulations on Hazardous Material Transport

- Comply with Tamil Nadu state guidelines on timing restrictions and permitted types of hazardous materials.
- Night travel might be restricted for hazardous materials, necessitating completion of transit during daylight.

Historical Incidents

- Few documented incidents involving heavy vehicles suggest a primarily safe route.
- Awareness and caution are crucial at intersections and pedestrian-heavy zones.

Environmental Considerations

- Travel through agricultural lands requires respecting local practices and avoiding contamination risks.
- Be mindful of wildlife crossings in rural areas and signs indicating sensitive zones.

Communication Coverage

- Major telecom providers serve the region well, but some rural patches might lack strong signals, especially in fields and less developed areas.
- Important to confirm communication equipment is functioning adequately before departure.

Estimated Emergency Response Times

- Average emergency response time can vary due to remoteness but generally could range from 30 minutes to an hour.
- Response is faster near towns with local police and medical facilities.

Overall Risk Assessment Summary

- **Medium Risk:** While largely safe, the route presents typical challenges of rural travel in India, including potential weather, roads, and minor communication issues.
- Vehicles should be well-maintained, drivers thoroughly briefed on local regulations, and contingency plans in place for unexpected situations or delays.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	Medium	11.43971, 77.87348	30 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	High	11.37868, 77.89498	15 KM/Hr
6	Turn	High	11.37850, 77.89453	15 KM/Hr
7	Turn	High	11.30859, 77.92918	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
7	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37868, 77.89498



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37850, 77.89453



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.30859, 77.92918

